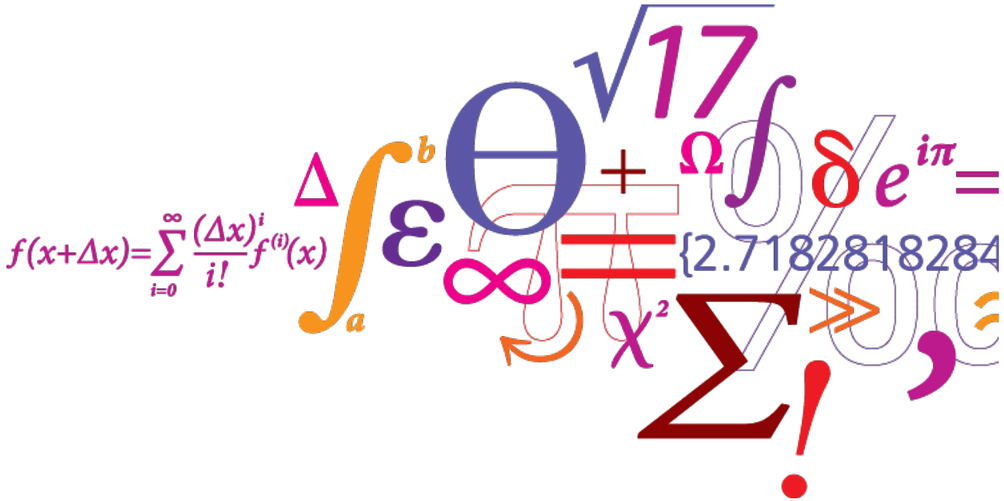
# 46755 – Renewables in Electricity Markets

## Lecture 9: Risk management

**Jalal Kazempour**

April 28, 2025



# Learning objectives of this lecture

After this lecture, you are expected be able to:

- Explain the concepts of **risk management**, **value-at-risk (VaR),** and **conditional value-at-risk (CVaR)**
- Formulate and solve a decision-making problem under uncertainty to derive optimal offers for a **risk-averse**, price-taking wind farm

# Recap

# Price-taking offering strategy for a wind farm

For simplicity, let's consider only the day-ahead and balancing markets, with both markets using the uniform pricing scheme.

**Input data:**

- Day-ahead wind power forecast per each time period (MW)
- Day-ahead market-clearing price forecast per each time period (€/MWh)
- Balancing market-clearing price forecast per each time period (€/MWh)

Offering strategy optimization problem

**(To be solved by a <u>price-taking</u> wind farm)**

Objective: Expected profit maximization

**Results:**

Production quantity offer (MW) in the day-ahead market (note that the offer price is assumed to be zero).

# Modeling uncertainty

Let's model uncertainty using a finite number of scenarios!

- Recall we have two sources of uncertainty: Wind power forecasts and day-ahead market prices in every time *t*.

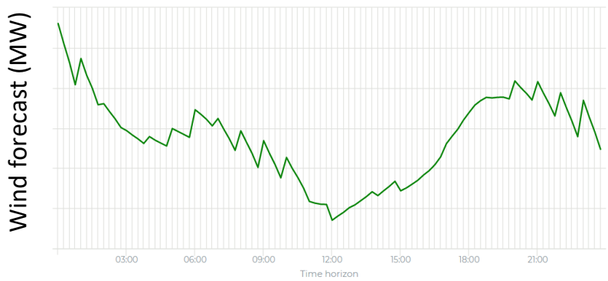
For example, assume there are ***two*** scenarios with the same probability (weight). In other words, the probability of each scenario is 0.5.

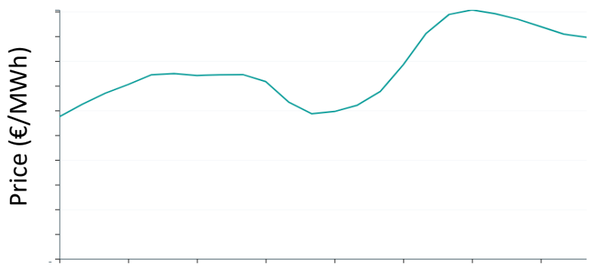
- Note that the sum of the probabilities of all scenarios should equal 1.

Hourly wind power forecast [MW]

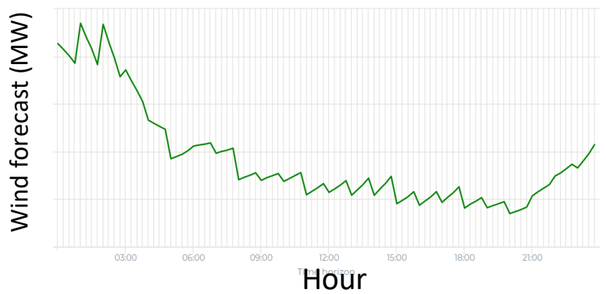
Hourly day-ahead market price forecast [€/MWh]

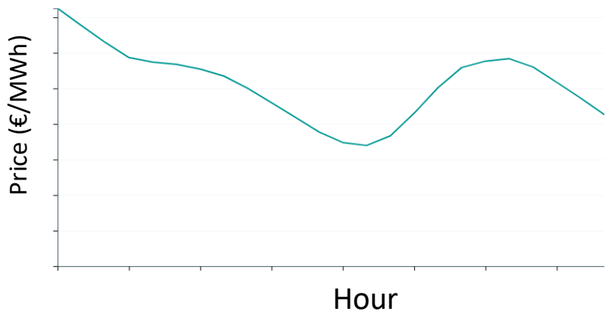
**Scenario 1**:




**Scenario 2**:



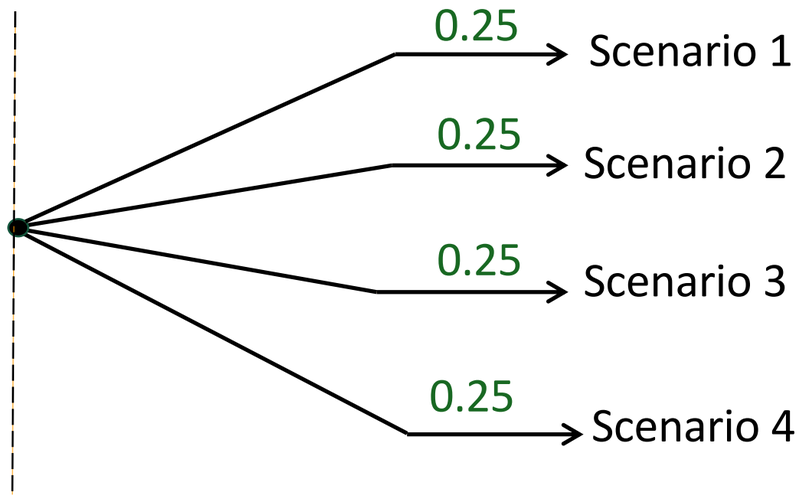
# Stochastic offering strategy problem

Imagine that in the day-ahead stage, based on the available forecast, we have 4 potential scenarios, each with identical probabilities:

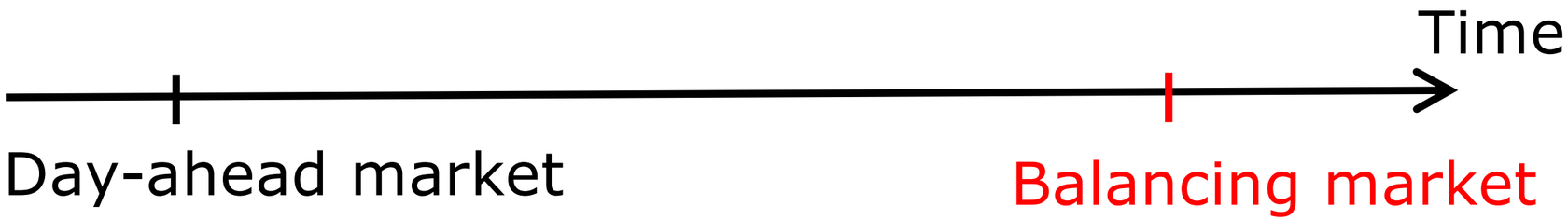
## Look-ahead strategy

Scenarios are generated based on the available information in the day-ahead stage, but they will only be realized in the balancing stage.

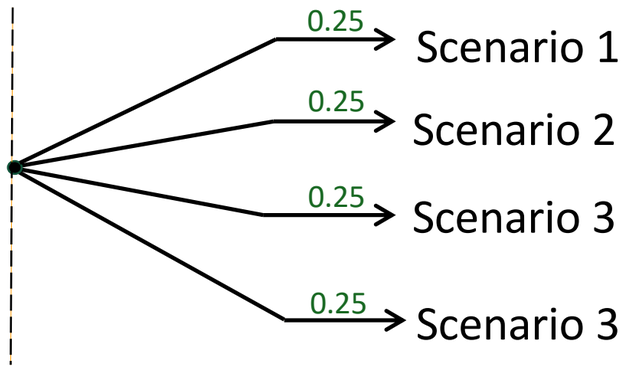
A decision must be made in the day-ahead stage, accounting for the potential scenarios in expectation.



# Stochastic offering strategy problem




Offering strategy based on stochastic forecasts, e.g.,




The true realization of wind power production

# Stochastic offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow}}{\text{Maximize}} \quad \sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right)$$

subject to:

$$0 \leq p_t^{\mathrm{DA}} \leq P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

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# Stochastic offering strategy problem

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subject to:

Set of time period (hours)

$$0 \leq p_t^{\mathrm{DA}} \leq P^{\mathrm{nom}} \quad \boxed{\forall t = \{1, 2, ..., 24\}}$$

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# Stochastic offering strategy problem

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subject to:

$$0 \leq p_t^{\mathrm{DA}} \leq P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

Set of scenarios

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\}, \boxed{\forall \omega = \{1, 2, ..., |\omega|\}}$$

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subject to:

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Number of scenarios (cardinality)

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., \boxed{|\omega|}\}$$

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Probability of scenario *w* (input data)

subject to:

$$0 \leq p_t^{\mathrm{DA}} \leq P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

Set of scenarios

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Probability of scenario *w* (input data)

Uncertain parameter: Forecasted day-ahead market price (€/MWh) in time *t* under scenario *w* (input data).

subject to:

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Set of scenarios

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \boxed{\forall \omega = \{1, 2, ..., |\omega|\}}$$

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Uncertain parameter: Wind power forecast (MW) in time *t* under scenario *w* (input data)

# Stochastic offering strategy problem

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All variables are indexed by scenario w, except for P^{DA}. Why?

subject to:

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**Answer:** Recall that in the day-ahead stage, we don't know which scenario will be realized. The wind farm should submit a single value to the day-ahead market (P^{DA}) as its production offer in each hour. Therefore, it cannot be scenario-indexed. However, considering the scenarios allows it to make a more informed (i.e., uncertainty-aware) offering decision!

# Stochastic offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow}}{\text{Maximize}} \quad \boxed{\sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right)}$$

Profit **in expectation** (weighted by probabilities)

subject to:

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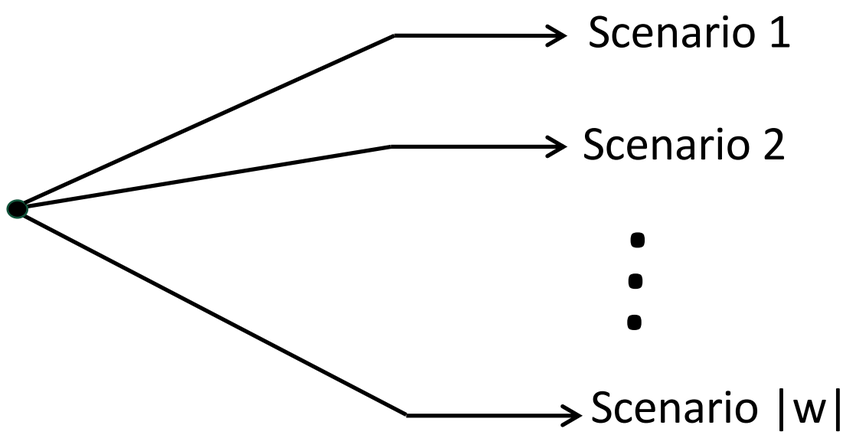
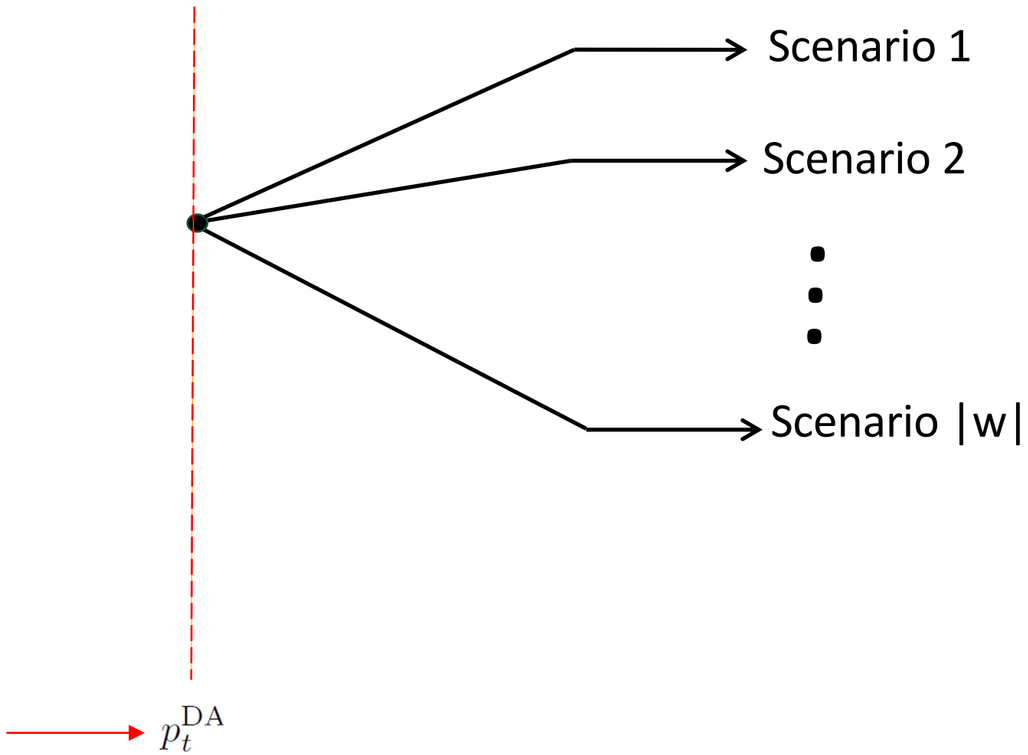
# Stochastic offering strategy problem



# Stochastic offering strategy problem

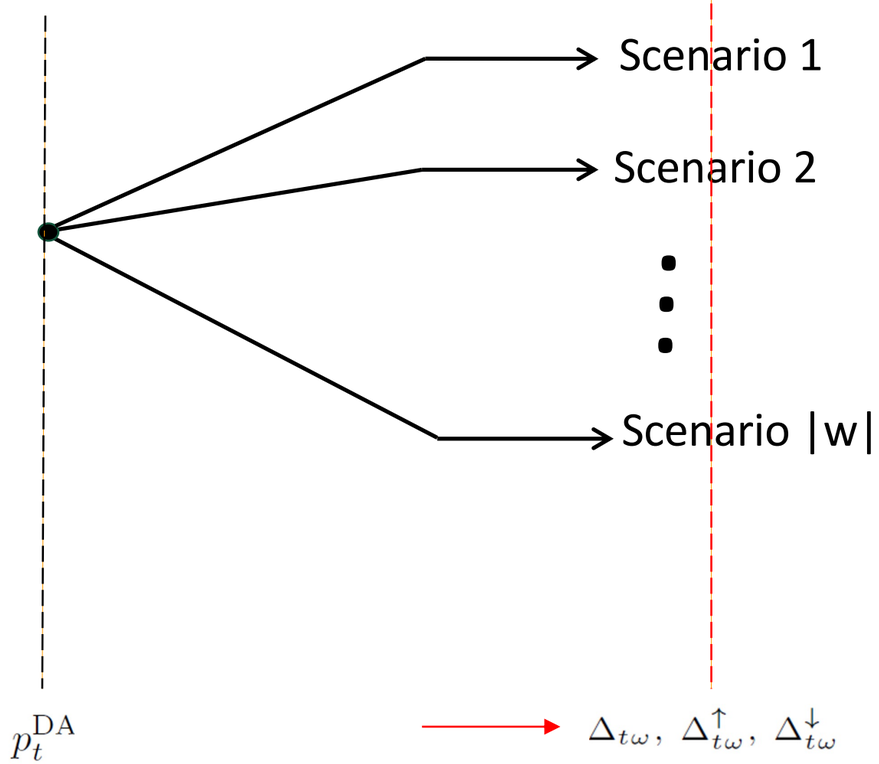
**First-stage** (here-an-now)
decisions (not indexed by scenario)



# Stochastic offering strategy problem

**First-stage** (here-an-now) decisions (not indexed by scenario)

**Second-stage** (wait-and-see) decisions (indexed by scenario)



# Risk management

DTU Wind, Technical University of Denmark

Jalal Kazempour

# Risk management

# Risk management

Notice that by incorporating multiple scenarios, the wind farm maximizes its expected (probability-weighted) profit:

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow}}{\text{Maximize}} \quad \underbrace{\sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right)}_{\textbf{Expected}\ \text{profit}}$$

# Risk management

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**Obviously, the profit under each individual scenario is not necessarily identical.**

**Example:**
Assume a wind farm whose expected profit over three equiprobable scenarios is €100. The distribution of profit across the scenarios could be:

- €200, €100, and €0 (a highly volatile distribution),
- or €150, €100, and €50 (a comparatively less volatile distribution),
- or any other combination.

# Risk management

Notice that by incorporating multiple scenarios, the wind farm maximizes its expected (probability-weighted) profit:

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- €200, €100, and €0 (a highly volatile distribution),
- or €150, €100, and €50 (a comparatively less volatile distribution),
- or any other combination.

A wind farm may be **risk-averse** (conservative) and prefer to reduce profit **volatility** across scenarios, even at the cost of a lower **expected** profit. By doing so, the wind farm protects itself against financially adverse outcomes in the tail of the distribution.

# Risk management

Notice that by incorporating multiple scenarios, the wind farm maximizes its expected (probability-weighted) profit:

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow}}{\text{Maximize}} \quad \underbrace{\sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right)}_{\textbf{Expected}\ \text{profit}}$$

While maintaining the underlying optimization problem as linear and convex, we can use a risk measure called "conditional value-at-risk (CVaR)" (also known as "expected shortfall") to model the wind farm's risk aversion.

# Risk management

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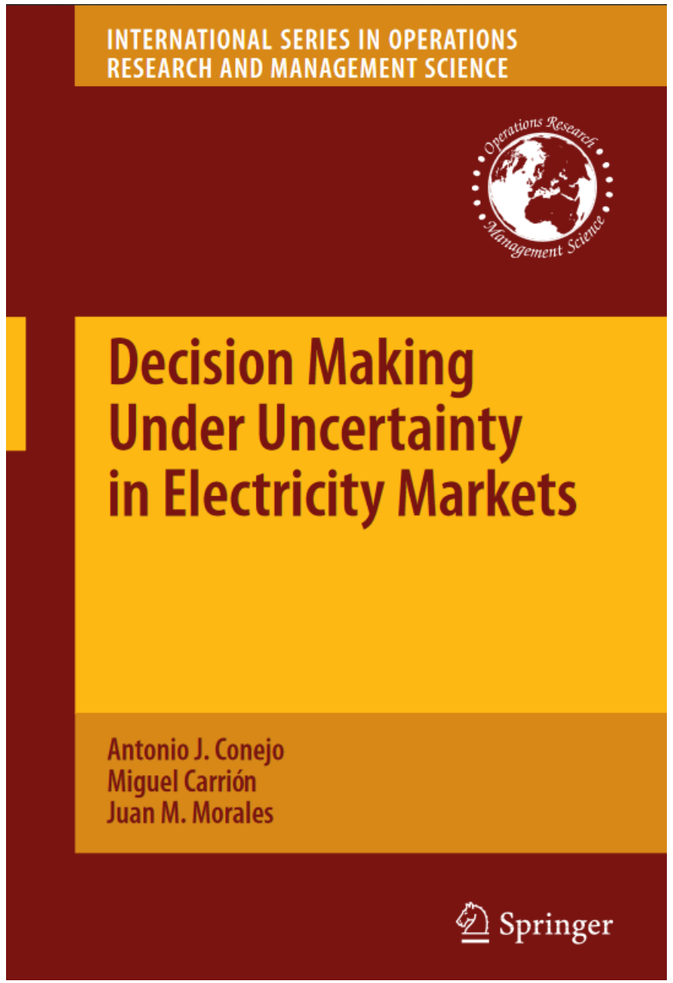
While maintaining the underlying optimization problem as linear and convex, we can use a risk measure called "conditional value-at-risk (CVaR)" (also known as "expected shortfall") to model the wind farm's risk aversion.

CVaR was proposed for the first time by **Rockafellar and Uryasev** [1]

[1] Rockafellar, R.T., Uryasev, S., "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, pp. 21–41, 2000

# Relevant reference




https://link.springer.com/book/10.1007/978-1-4419-7421-1

Chapter 4: Risk Management

# Conditional value-at-risk (CVaR)

Although we use a finite set of scenarios, let's for now assume that the wind farm's profit follows an underlying **continuous** probability distribution.

- The Probability Density Function (PDF) of the wind farm's profit might look as follows:

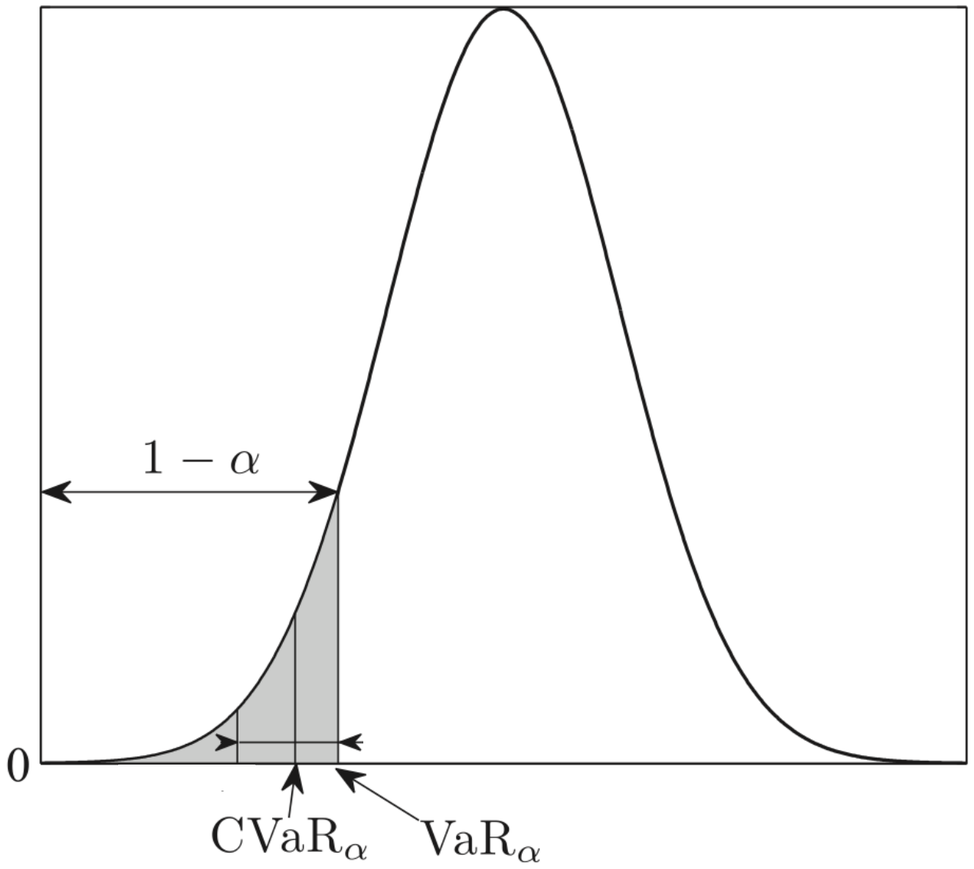
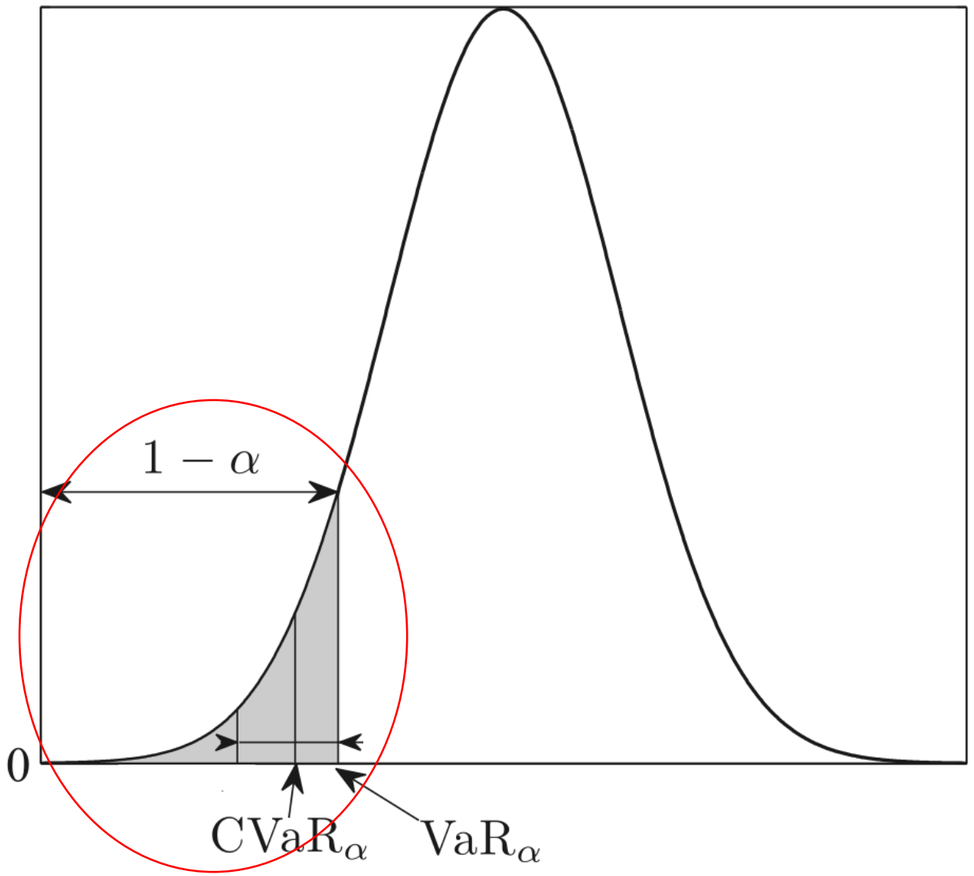



Figure borrowed from: https://doi.org/10.1016/j.eneco.2012.03.016

# Conditional value-at-risk (CVaR)

The Probability Density Function (PDF) of the wind farm's profit might look as follows:

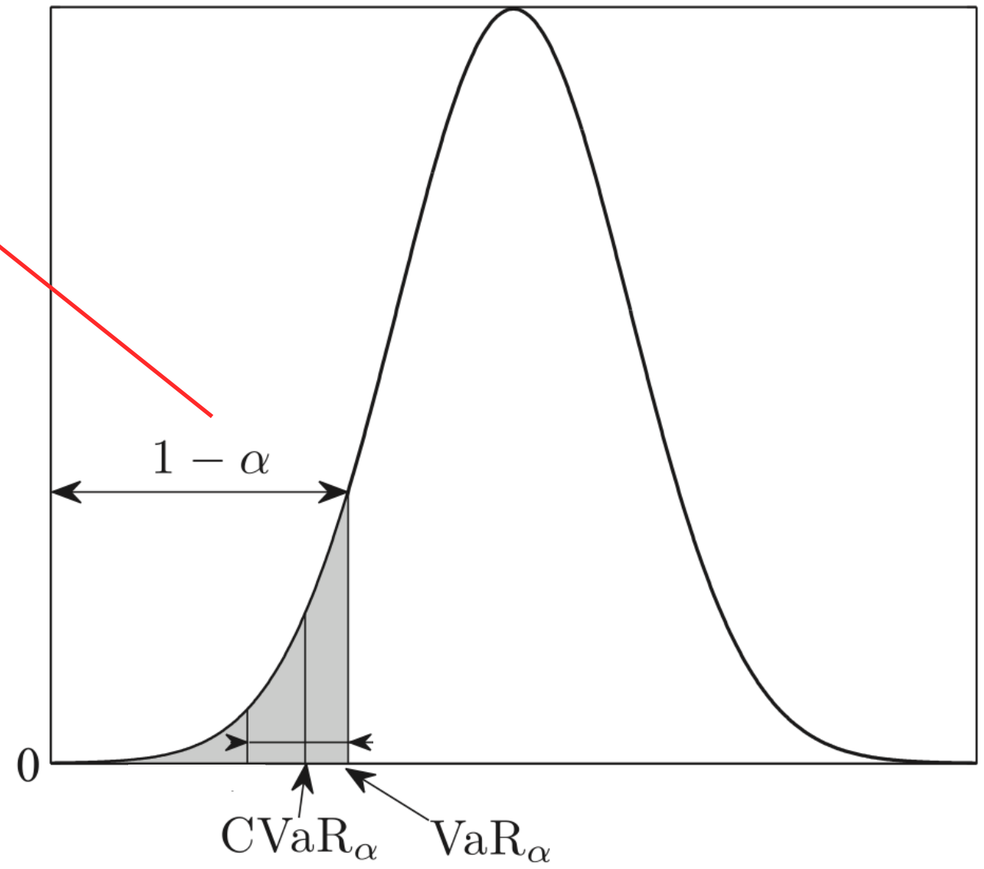
The **lower-profit tail** of the distribution, which the wind farm operator might be particularly concerned about, especially if it is a "fat" tail.



# Conditional value-at-risk (CVaR)

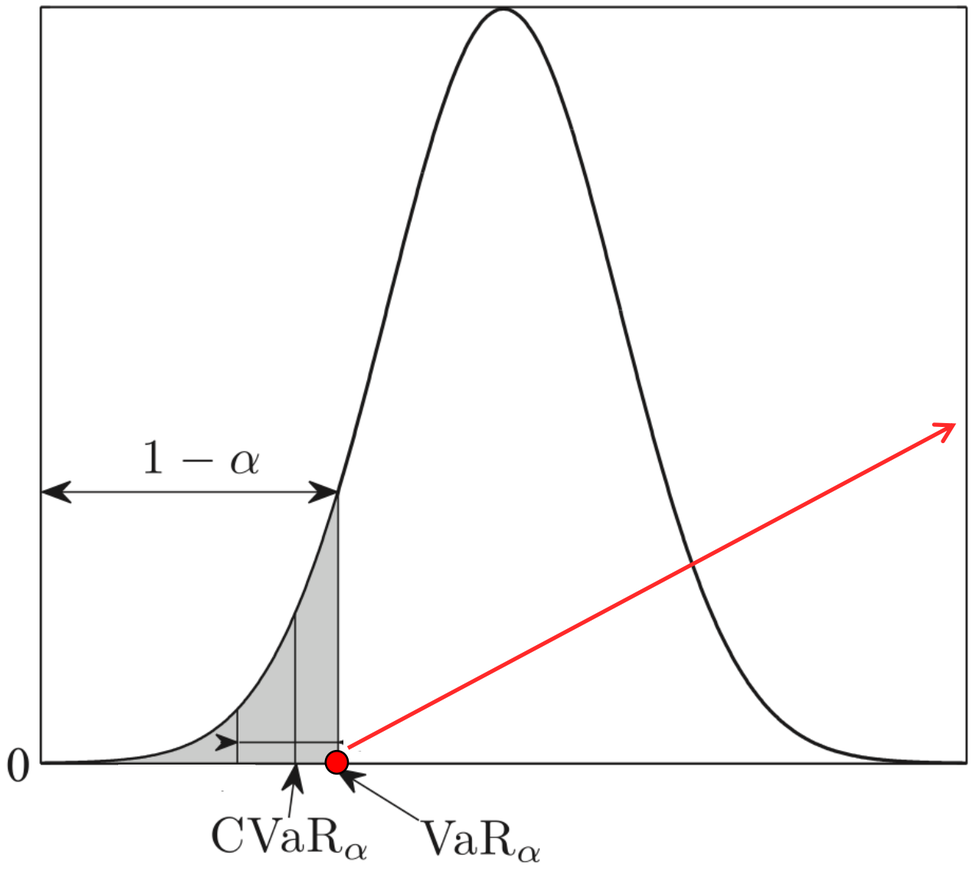
The Probability Density Function (PDF) of the wind farm's profit might look as follows:

(1−α)-quantile of the distribution, where 0 ≤ α ≤ 1



# Conditional value-at-risk (CVaR)

The Probability Density Function (PDF) of the wind farm's profit might look as follows:



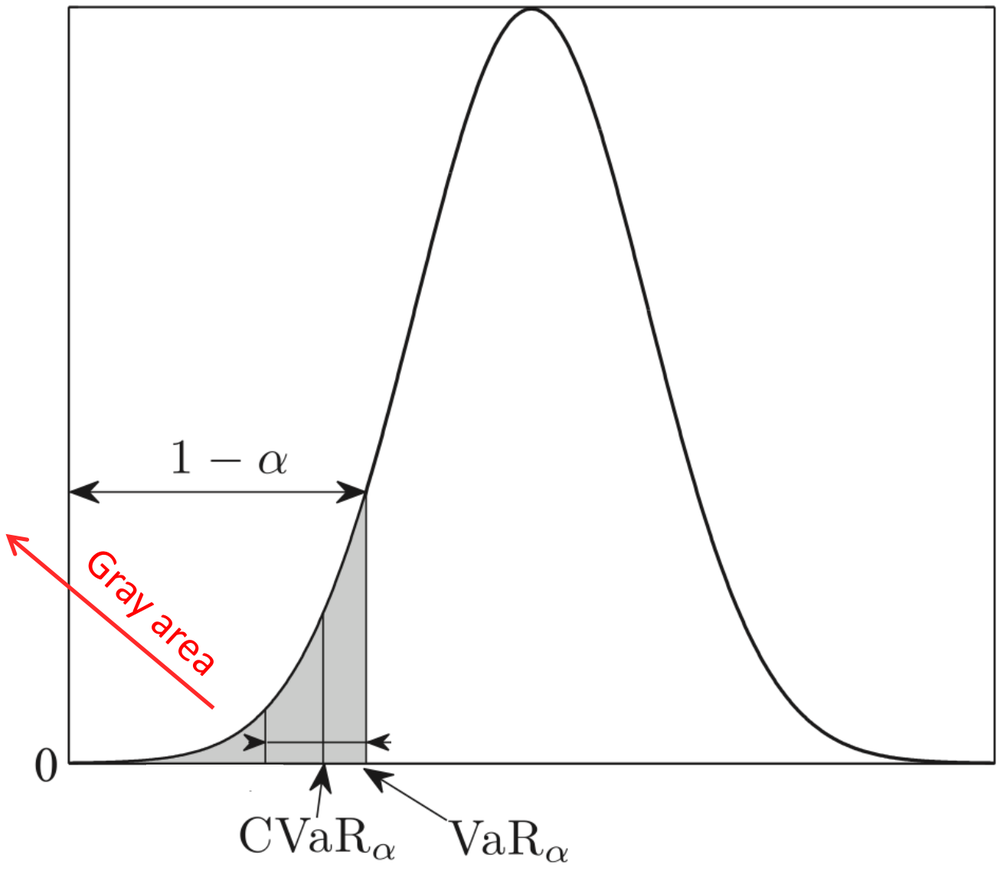

**VaR (Value-at-Risk)** is the minimum profit such that the probability of the profit being lower than this value is less than or equal to 1−α.

# Conditional value-at-risk (CVaR)

The Probability Density Function (PDF) of the wind farm's profit might look as follows:

**CVaR (Conditional Value-at-Risk)** is the expected profit that is lower than the (1−α)-quantile of the profit distribution.

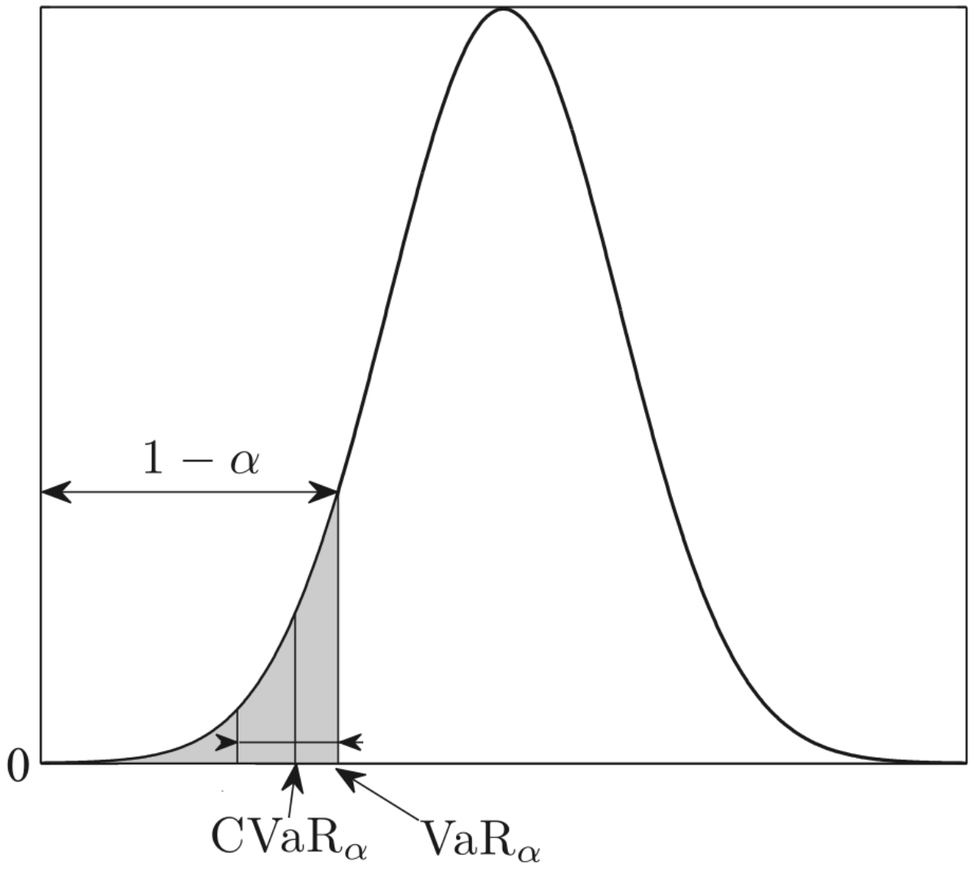
- In other words, CVaR is the expected value in the tail region where profit is lower than the VaR.



# Conditional value-at-risk (CVaR)

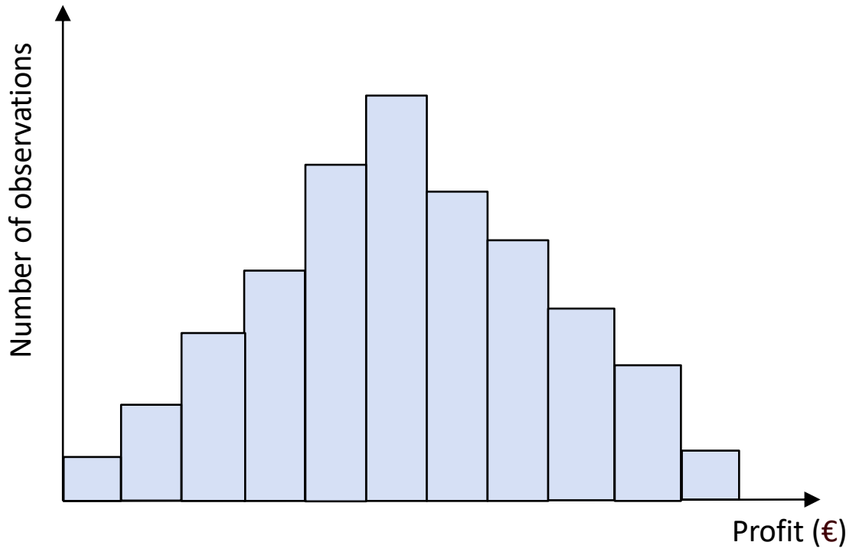
The Probability Density Function (PDF) of the wind farm's profit might look as follows:

**Note:** Unlike VaR, CVaR is a key risk measure when the underlying distribution has a 'fat tail' (i.e., a larger tail region, or gray area, where extreme losses occur).



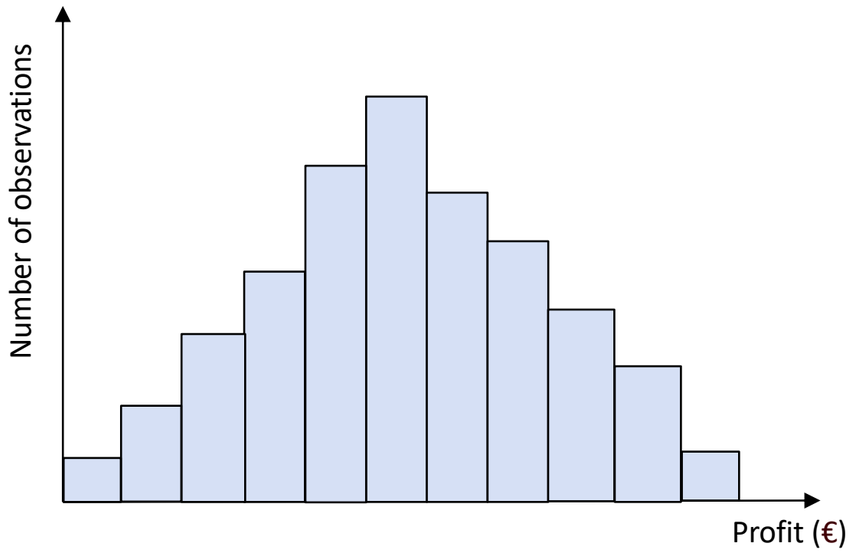
# Conditional value-at-risk (CVaR)

Let's now consider a **discrete** distribution for wind farm's profit:



# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **expected profit**:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow}}{\text{Maximize}} \quad \sum_{\omega} \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)$$

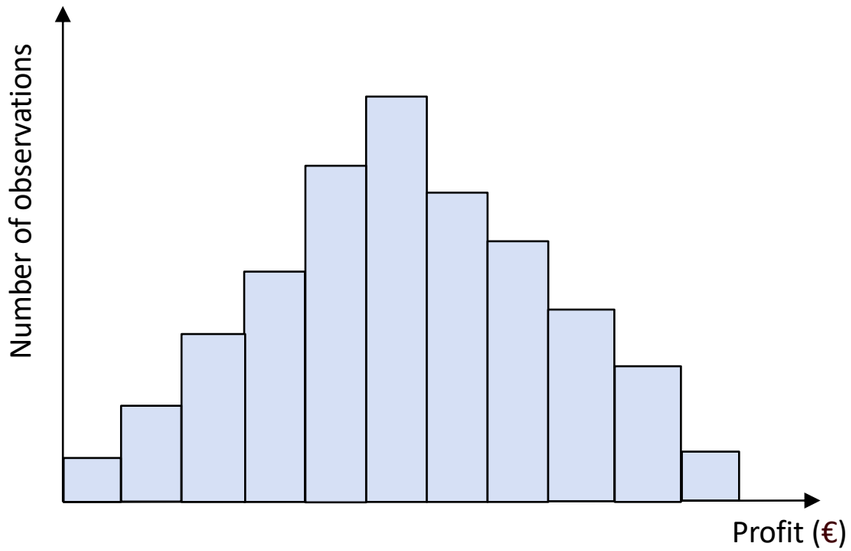
subject to:

$$f\left(P^{\mathrm{DA}}\right) \ \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \ \leq 0 \quad \forall \omega$$

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:

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$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow}}{\text{Maximize}} \quad \sum_{\omega} \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)$$

Day-ahead offer (MW)

Imbalances under scenario *w* (MW)

subject to:

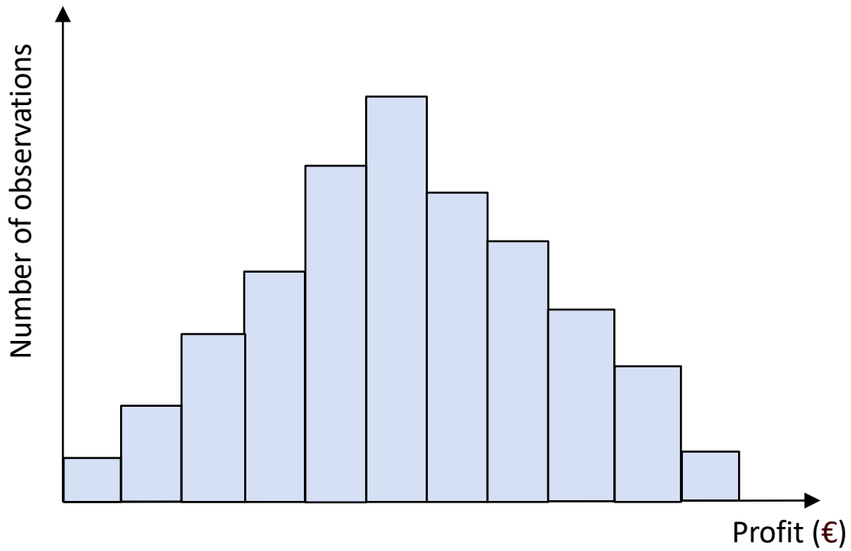
$$f\left(P^{\mathrm{DA}}\right) \ \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \ \leq 0 \quad \forall \omega$$

Technical constraints

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **expected profit**:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow}}{\text{Maximize}} \quad \sum_{\omega} \underbrace{\pi_\omega}_{\text{Probability of scenario } w} \underbrace{\text{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)}_{\text{Profit function under scenario } w \text{ (€)}}$$

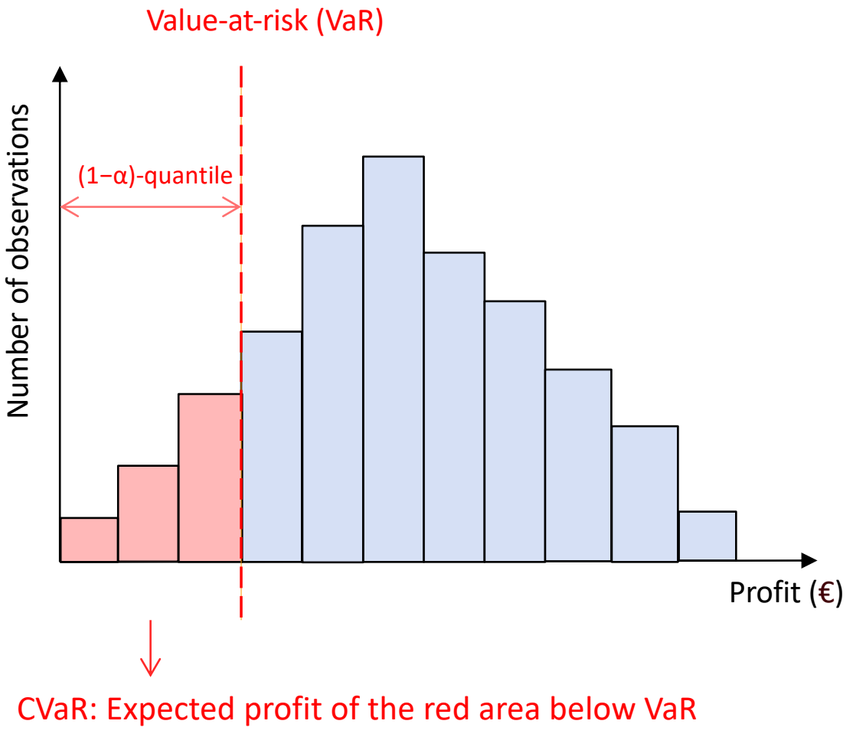
$$\text{subject to:}$$

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall \omega$$

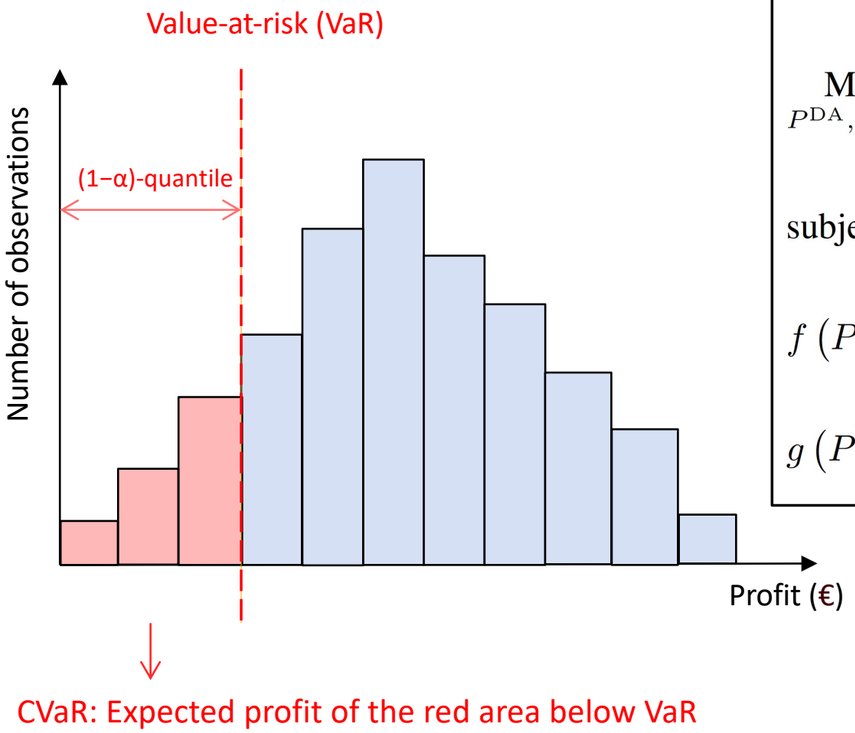
# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:



# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \ \max\left[0, \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]$$

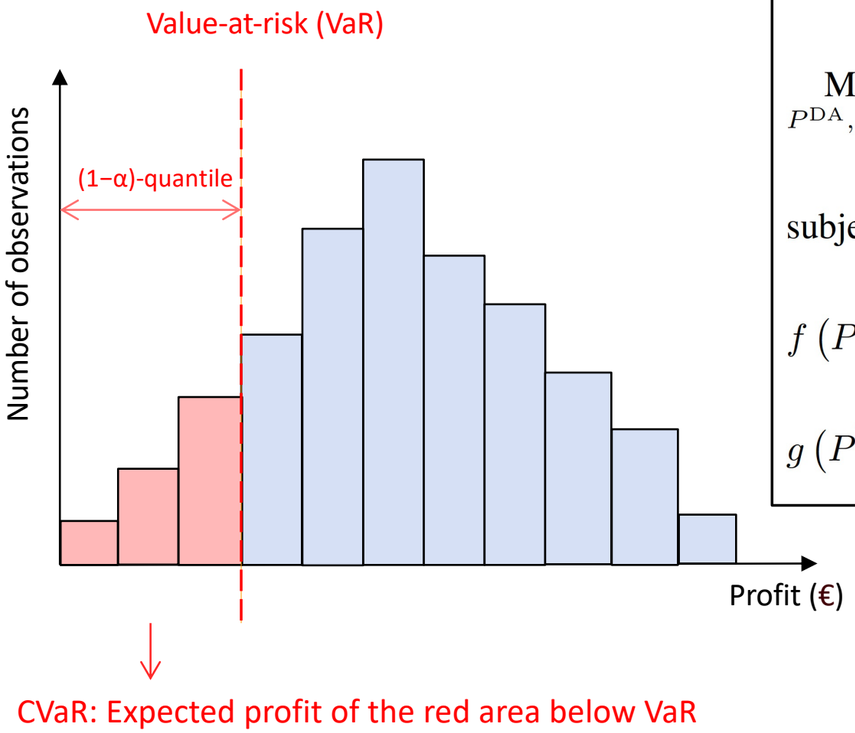
subject to:

$$f\left(P^{\mathrm{DA}}\right) \ \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \ \leq 0 \quad \forall\omega$$

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

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subject to:

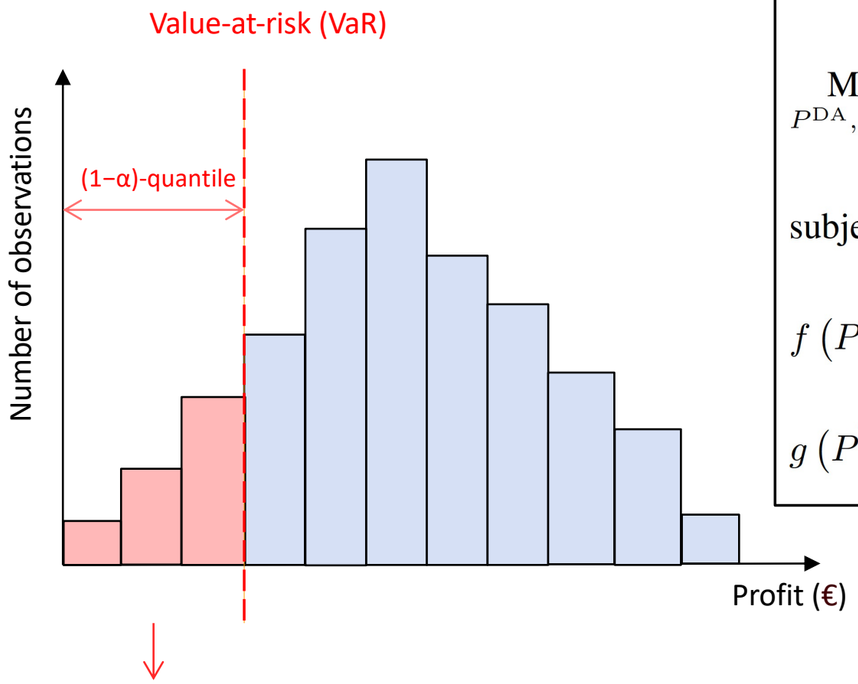
$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

VaR (€)
(it is a variable to be detremined by the optimization problem)

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \max\left[0, \zeta - \text{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]$$

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

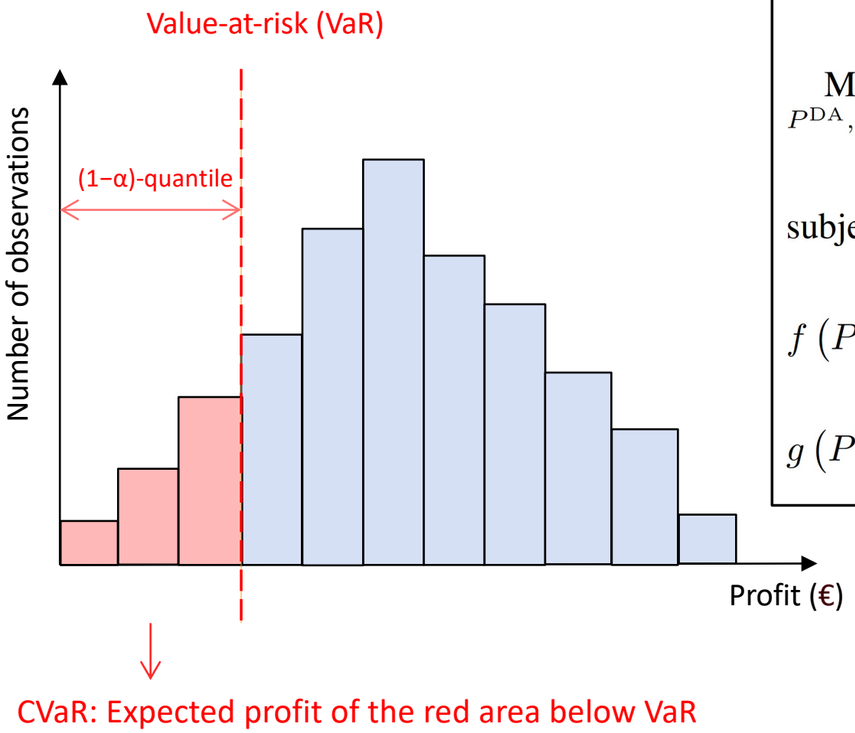
$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

VaR (€)
(it is a variable to be detremined by the optimization problem)

Note that the optimal value of VaR is determined endogenously (it is <u>not</u> a parameter).

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^\uparrow,\ \Delta_\omega^\downarrow,\ \zeta}{\text{Maximize}} \quad \underbrace{\zeta - \frac{1}{1-\alpha}\sum_\omega \pi_\omega \max\left[0, \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^\uparrow, \Delta_\omega^\downarrow\right)\right]}_{\text{The definition of CVaR (€) according to Rockafellar and Uryasev [1]}}$$

subject to:

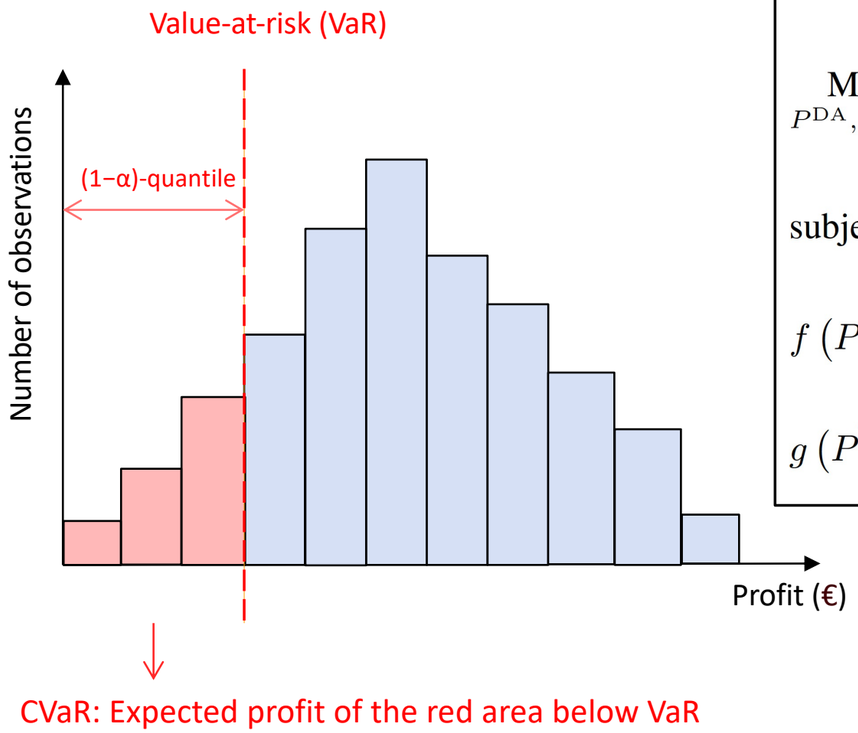
$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^\uparrow, \Delta_\omega^\downarrow\right) \leq 0 \quad \forall \omega$$

[1] Rockafellar, R.T., Uryasev, S., "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, pp. 21–41, 2000

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \ \max\left[0, \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]$$

subject to:

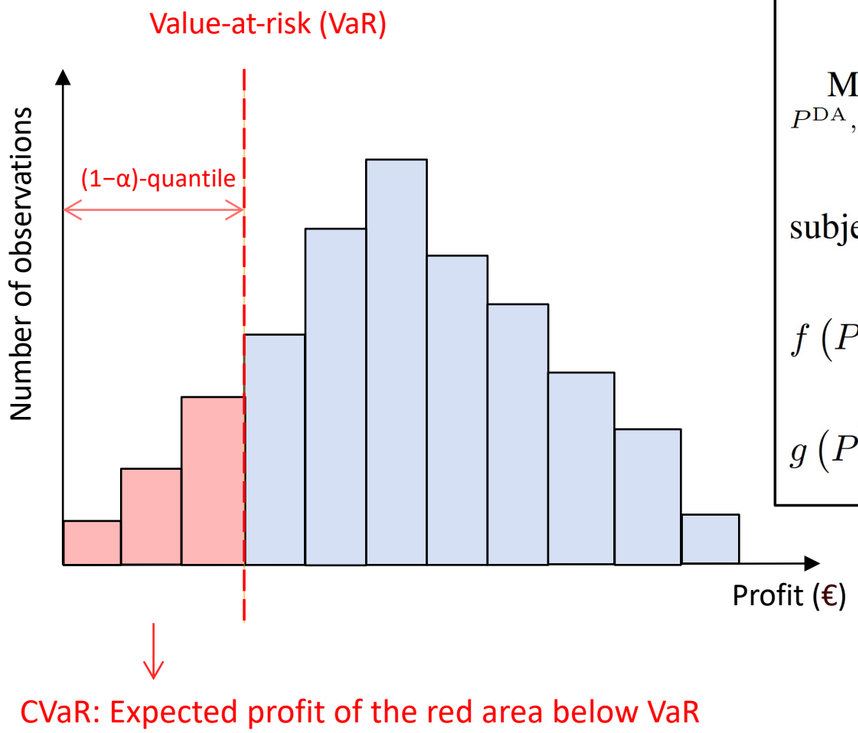
$$f\left(P^{\mathrm{DA}}\right) \ \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \ \leq 0 \quad \forall\omega$$

How to reformulate the term **max[.,.]** in the objective functiom?

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




Maximization of the **CVaR:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \ \max\left[0, \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]$$

subject to:

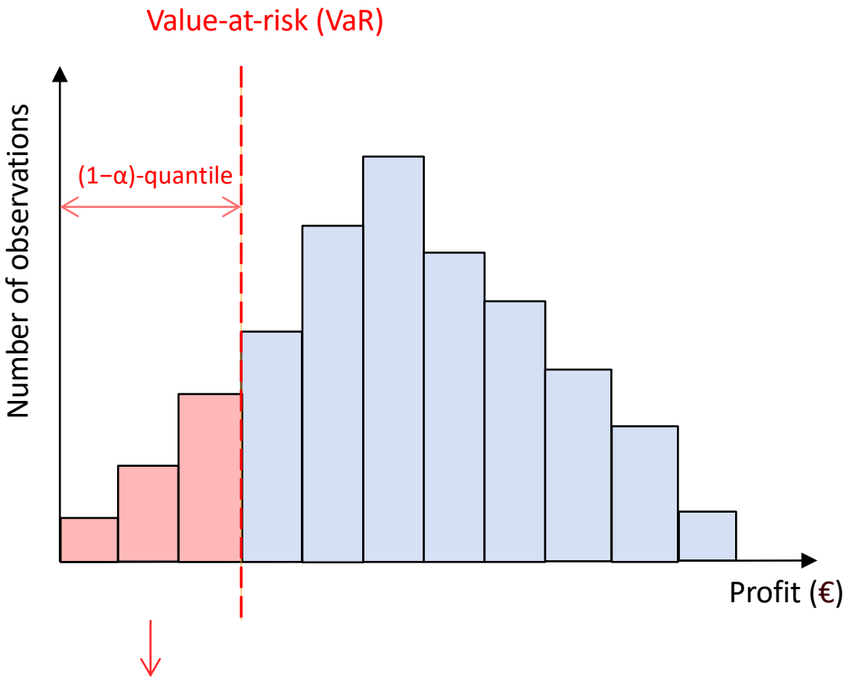
$$f\left(P^{\mathrm{DA}}\right) \ \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \ \leq 0 \quad \forall\omega$$

How to reformulate the term **max[.,.]** in the objective functiom?
By defining an auxiliary variable!

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega\eta_\omega$$

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

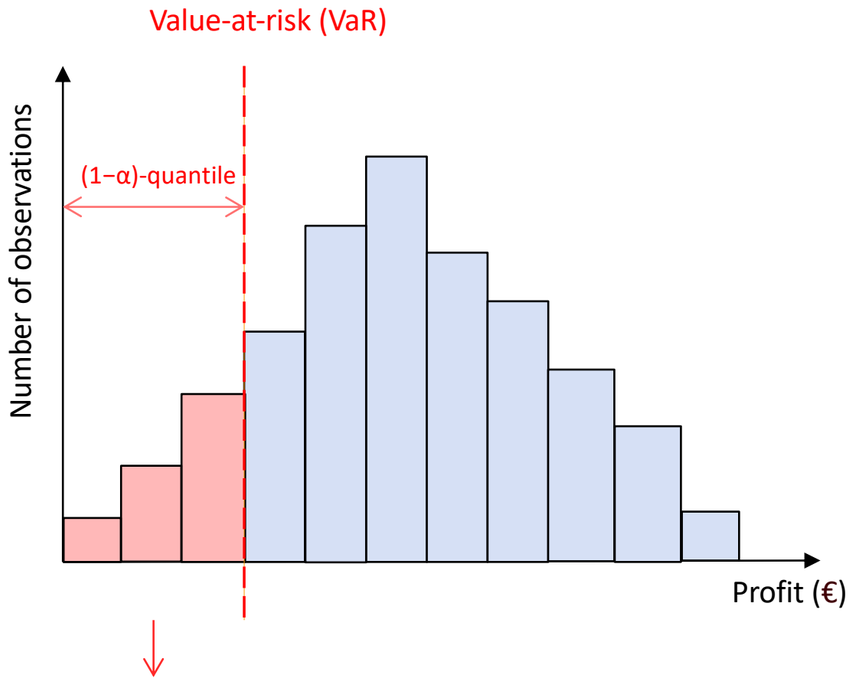
$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

$$\eta_\omega \geq 0 \quad \forall\omega$$

$$\eta_\omega \geq \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \quad \forall\omega$$

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \eta_\omega$$

An auxiliary variable (€)

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

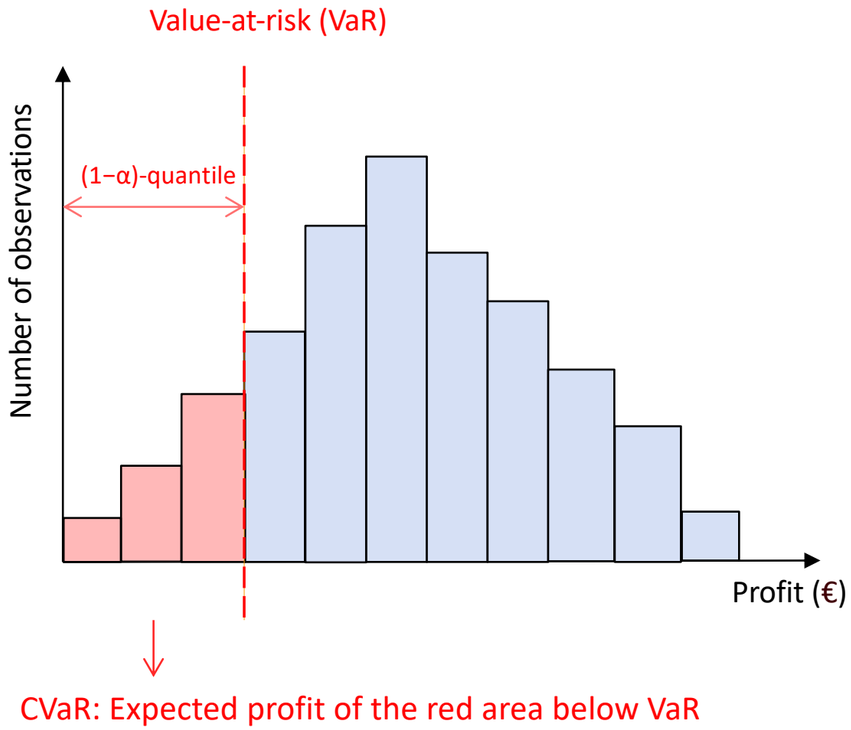
$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

$$\eta_\omega \geq 0 \quad \forall\omega$$

$$\eta_\omega \geq \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \quad \forall\omega$$

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:




**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \underbrace{\zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \eta_\omega}_{\text{CVaR (€)}}$$

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

$$\eta_\omega \geq 0 \quad \forall\omega$$

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# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:

**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \underbrace{\zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega\eta_\omega}_{\text{CVaR (€)}}$$

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

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$$\eta_\omega \geq 0 \quad \forall\omega$$

$$\eta_\omega \geq \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \quad \forall\omega$$

**Note 1/3:**

The CVaR formulation preserves the **linearity** of the original optimization problem.

Therefore, if the original expected profit maximization problem is linear, the CVaR maximization problem will also remain linear.

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:

**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \underbrace{\zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \eta_\omega}_{\text{CVaR (€)}}$$

subject to:

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$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

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$$\eta_\omega \geq \zeta - \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \quad \forall\omega$$

**Note 2/3:**

The dual variable of this constraint can be interpreted as "**risk-adjusted probabilities**", in contrast to statistical probabilities $\pi_\omega$ [2].

[2] A. Ehrenmann and Y. Smeers, "Generation capacity expansion in a risky environment: A stochastic equilibrium analysis," *Operations Research*, vol. 59, no. 6, pp. 1332–1346, 2011.

# Conditional value-at-risk (CVaR)

Let's now consider a discrete distribution for wind farm's profit:

**Note 3/3:**

If all scenarios are equiprobable (i.e., each with the same probability), CVaR is equal to the expected profit in the worst (1−α) × 100% of scenarios.

**Reformulation** of the **CVaR-maximiation** problem:

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \underbrace{\zeta - \frac{1}{1-\alpha}\sum_{\omega}\pi_\omega \eta_\omega}_{\text{CVaR (€)}}$$

subject to:

$$f\left(P^{\mathrm{DA}}\right) \leq 0$$

$$g\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \leq 0 \quad \forall\omega$$

$$\eta_\omega \geq 0 \quad \forall\omega$$

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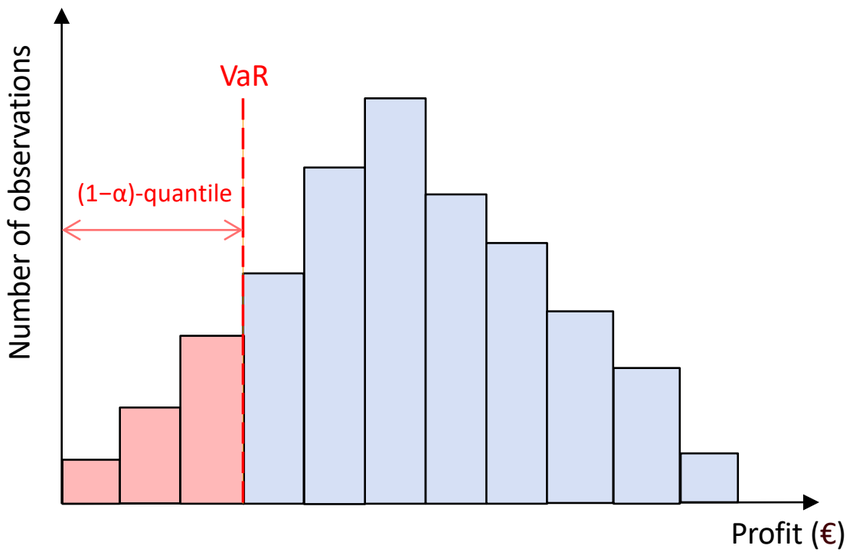
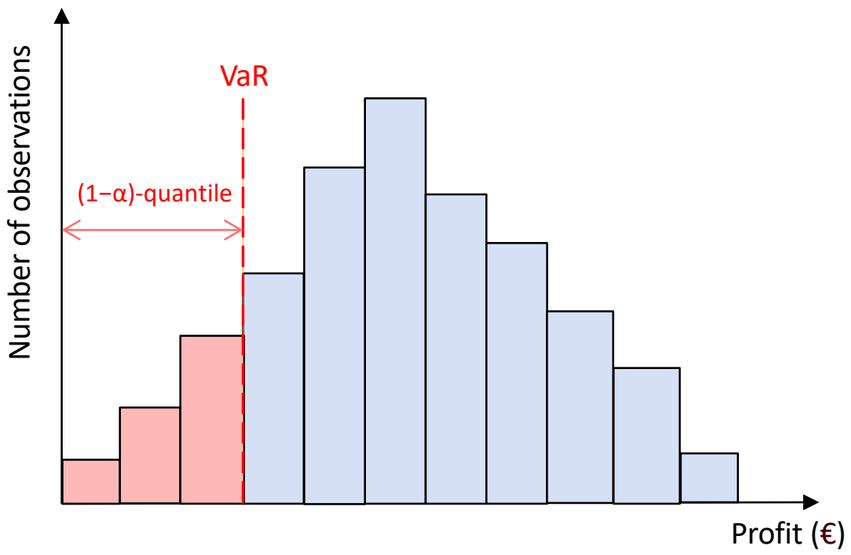
# Expected profit vs. CVaR



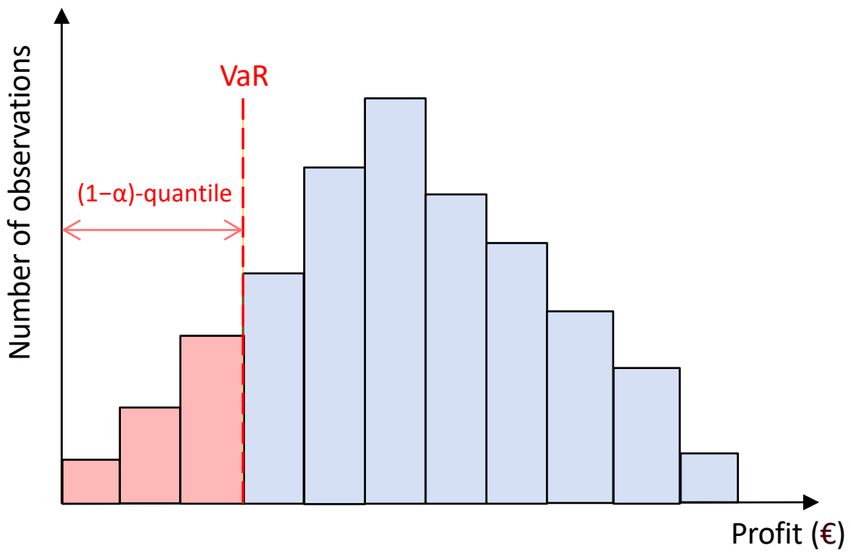

Should we maximize expected profit or CVaR? Is there a way to consider both?

# Expected profit vs. CVaR




Should we maximize expected profit or CVaR? Is there a way to consider both? Yes, let's formulate a multi-objective optimization problem!

# Expected profit vs. CVaR

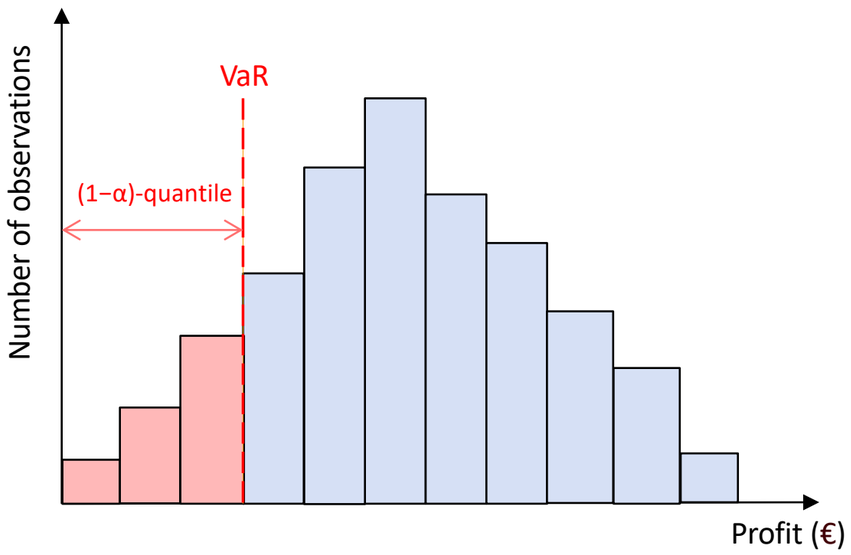



Should we maximize expected profit or CVaR? Is there a way to consider both? Yes, let's formulate a multi-objective optimization problem!

**Revised objective function:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \left[ \sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left( P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow} \right) \right] + \beta \left[ \zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega \right]$$

# Expected profit vs. CVaR

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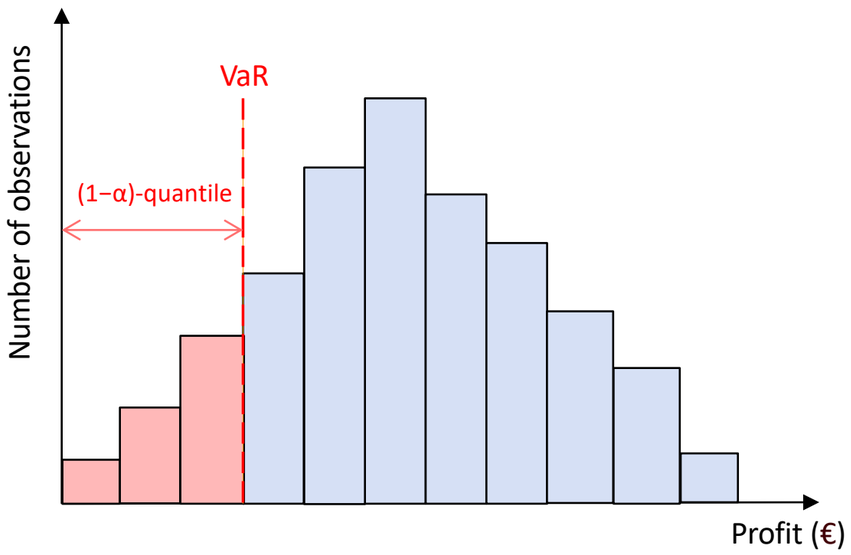
$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \underbrace{\left[\sum_\omega \pi_\omega\ \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]}_{\text{Expected profit (€)}} + \beta \underbrace{\left[\zeta - \frac{1}{1-\alpha}\sum_\omega \pi_\omega \eta_\omega\right]}_{\text{CVaR (€)}}$$

# Expected profit vs. CVaR




Should we maximize expected profit or CVaR? Is there a way to consider both? Yes, let's formulate a multi-objective optimization problem!

**Given non-negative weight (parameter)**

**Revised objective function:**

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \underbrace{\left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega\left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right]}_{\text{Expected profit (€)}} + \beta \underbrace{\left[\zeta - \frac{1}{1-\alpha}\sum_\omega \pi_\omega \eta_\omega\right]}_{\text{CVaR (€)}}$$

# Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \left[ \sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left( P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow} \right) \right] + \beta \left[ \zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega \right]$$

# Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \left[ \sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \right] + \beta \left[ \zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega \right]$$

2) For $0 \leq \beta \leq 1$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} (1-\beta) \left[ \sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right) \right] + \beta \left[ \zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega \right]$$

# Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

2) For $0 \leq \beta \leq 1$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad (1-\beta)\left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

**Both versions of the weighted objectve function can be used!**

## Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

2) For $0 \leq \beta \leq 1$

$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad (1-\beta)\left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

In both, $\beta = 0$ implies a **risk-neutral** case (expected profit maximizationn only).

# Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

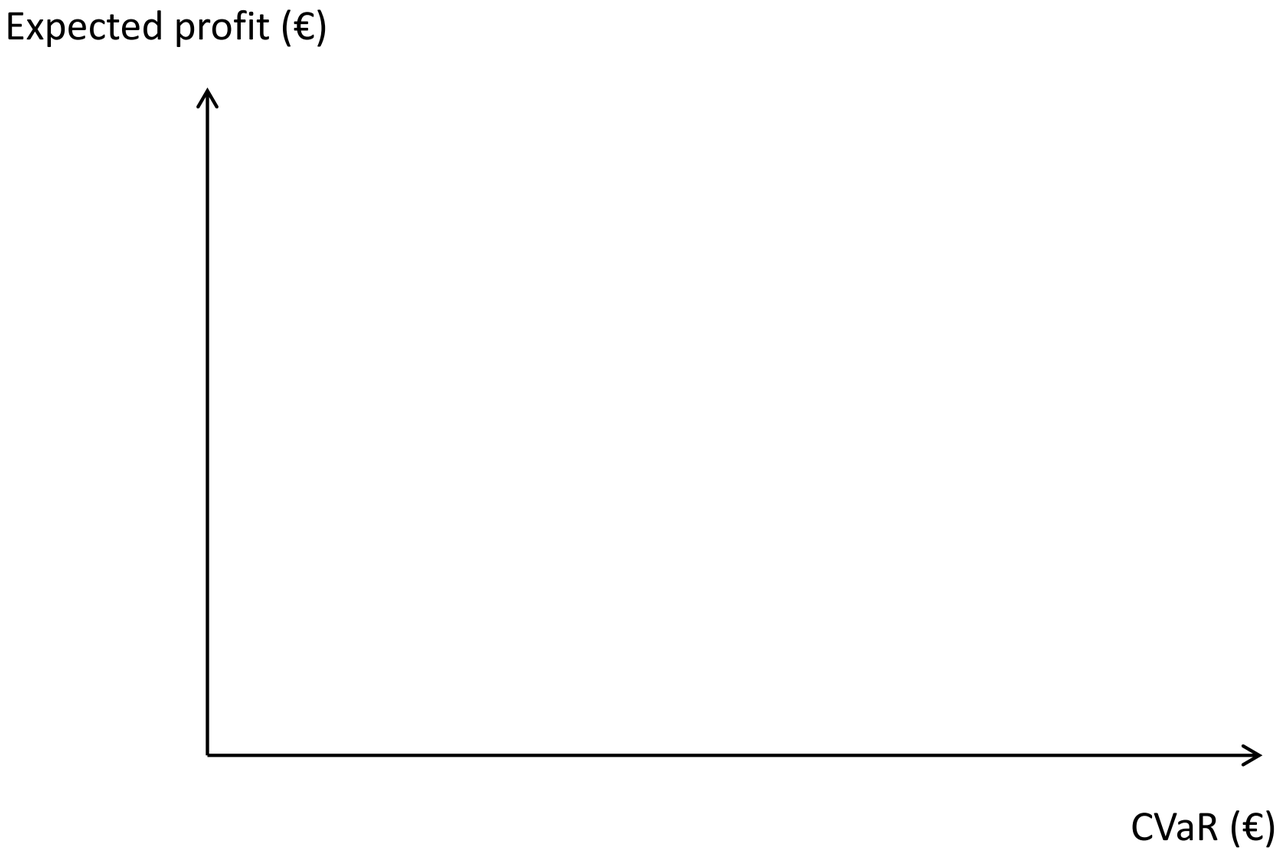
$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

2) For $0 \leq \beta \leq 1$

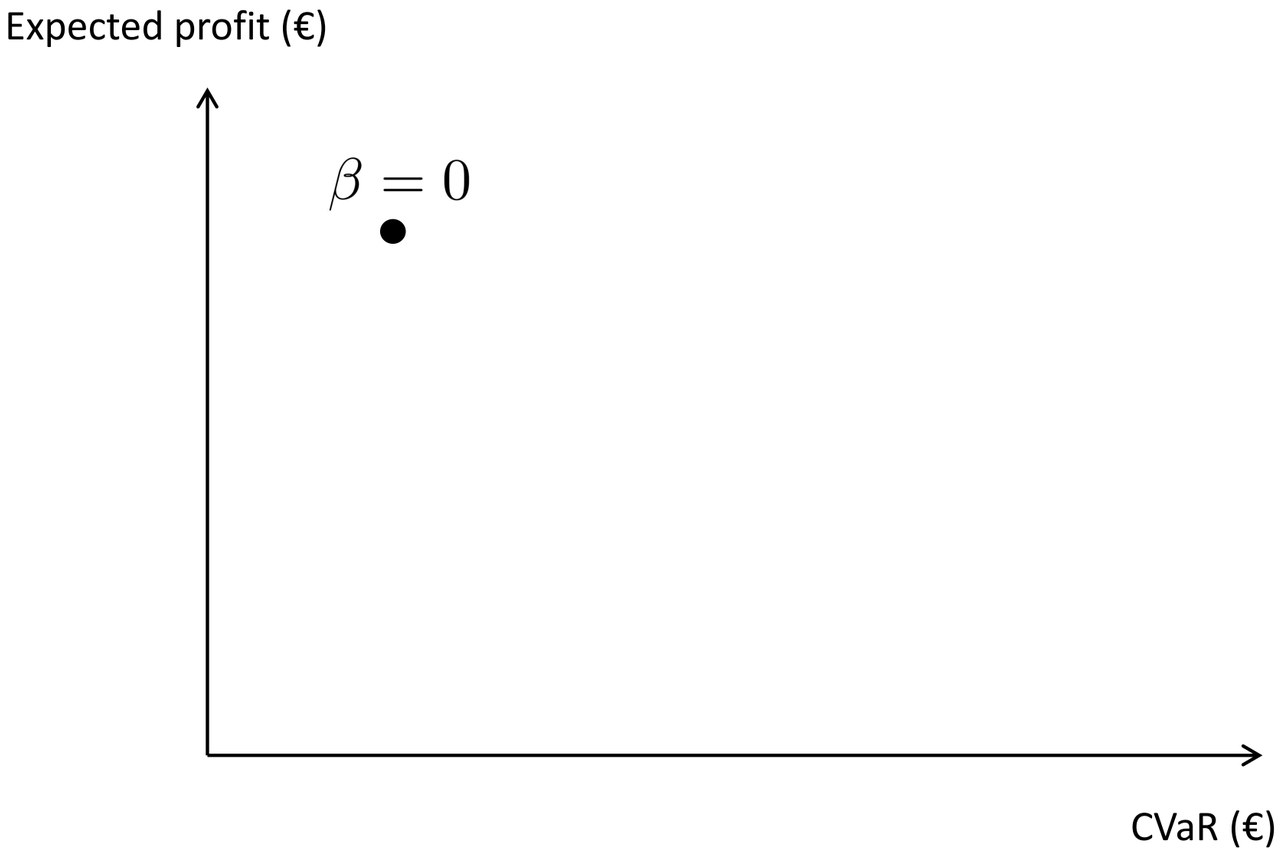
$$\underset{P^{\mathrm{DA}},\ \Delta_\omega^{\uparrow},\ \Delta_\omega^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad (1-\beta)\left[\sum_\omega \pi_\omega \ \mathrm{Profit}_\omega \left(P^{\mathrm{DA}}, \Delta_\omega^{\uparrow}, \Delta_\omega^{\downarrow}\right)\right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_\omega \pi_\omega \eta_\omega\right]$$

In both, higher values of $\beta$ imply greater **conservativeness**.

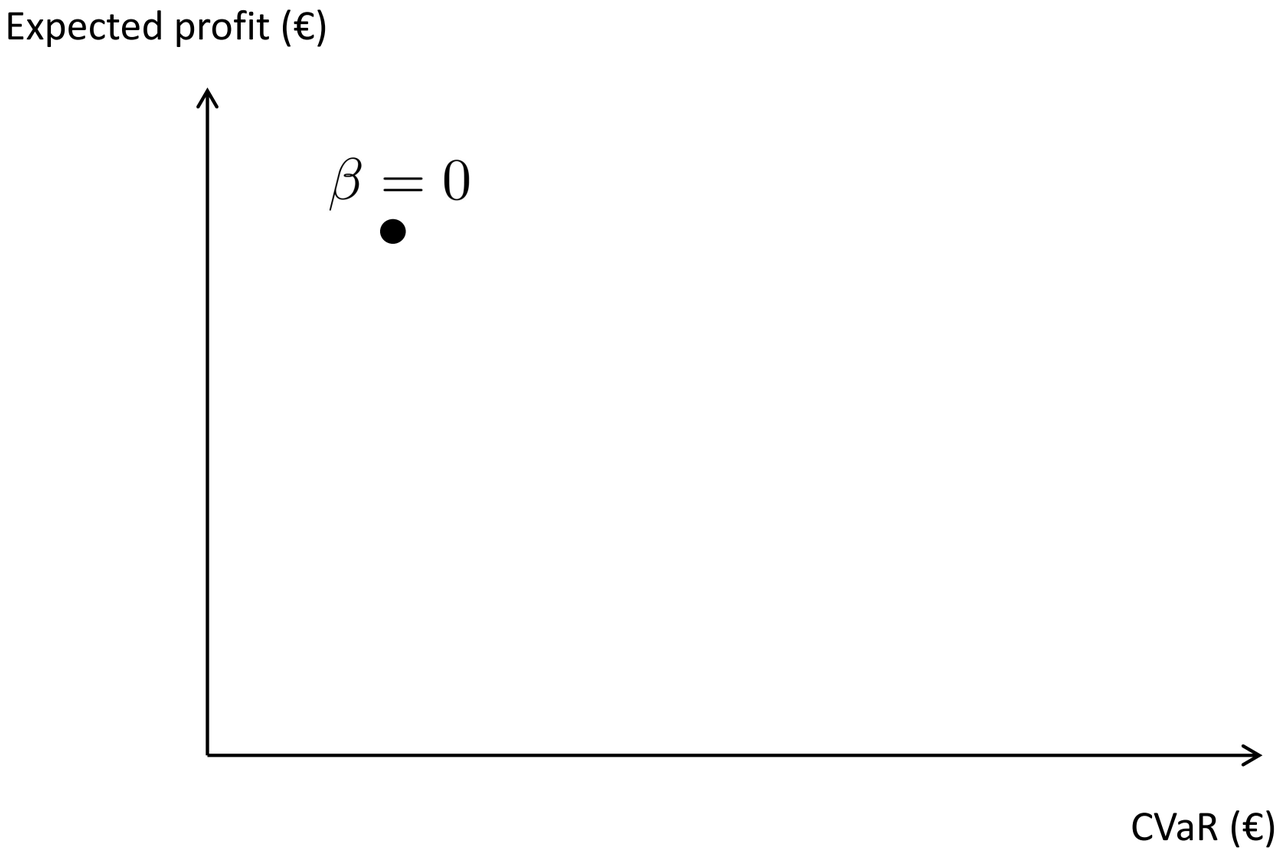
# Efficient frontier



# Efficient frontier



# Efficient frontier




Question: How will the curve look like as β increases?

# Risk-averse offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \left(\sum_{t=1}^{24}\sum_{\omega=1}^{|\omega|}\pi_\omega\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right)\right) + \beta\left(\zeta - \frac{1}{1-\alpha}\sum_{\omega=1}^{|\omega|}\pi_\omega\ \eta_\omega\right)$$

subject to:

$$0 \le p_t^{\mathrm{DA}} \le P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\uparrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\downarrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\eta_\omega \ge 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

$$-\sum_{t=1}^{24}\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right) + \zeta - \eta_\omega \le 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

# Risk-averse offering strategy problem

New additions (all other parts remain unchanged):

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow},\ \boxed{\zeta,\ \eta_\omega}}{\text{Maximize}} \quad \left(\sum_{t=1}^{24}\sum_{\omega=1}^{|\omega|}\pi_\omega\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right)\right) + \beta\boxed{\left(\zeta - \frac{1}{1-\alpha}\sum_{\omega=1}^{|\omega|}\pi_\omega\ \eta_\omega\right)}$$

subject to:

$$0 \le p_t^{\mathrm{DA}} \le P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\uparrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\downarrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\eta_\omega \ge 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

$$-\sum_{t=1}^{24}\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right) + \zeta - \eta_\omega \le 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

# Risk-averse offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow},\ \zeta,\ \eta_\omega}{\text{Maximize}} \quad \left(\sum_{t=1}^{24}\sum_{\omega=1}^{|\omega|}\pi_\omega\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right)\right) + \beta\left(\zeta - \frac{1}{1-\alpha}\sum_{\omega=1}^{|\omega|}\pi_\omega\ \eta_\omega\right)$$

VaR (not indexed by scenario)

subject to:

$$0 \le p_t^{\mathrm{DA}} \le P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\uparrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

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$$\eta_\omega \ge 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

$$-\sum_{t=1}^{24}\left(\lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow}\right) + \zeta - \eta_\omega \le 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

# Risk-averse offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow},\ \zeta,\ \boxed{\eta_\omega}}{\text{Maximize}} \left( \sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right) \right) + \beta \boxed{\left( \zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{|\omega|} \pi_\omega\ \eta_\omega \right)}$$

auxiliary variable (scenario-indexed)

subject to:

$$0 \leq p_t^{\mathrm{DA}} \leq P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\uparrow} \geq 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

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$$\eta_\omega \geq 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

$$-\sum_{t=1}^{24} \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right) + \zeta - \eta_\omega \leq 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

# Risk-averse offering strategy problem

$$\underset{p_t^{\mathrm{DA}},\ \Delta_{t\omega},\ \Delta_{t\omega}^{\uparrow},\ \Delta_{t\omega}^{\downarrow},\ \boxed{\zeta,\ \eta_\omega}}{\text{Maximize}} \left( \sum_{t=1}^{24} \sum_{\omega=1}^{|\omega|} \pi_\omega \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right) \right) + \beta \boxed{\left( \zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{|\omega|} \pi_\omega\ \eta_\omega \right)}$$

**CVaR term:** The expected profit of the wind farm in scenarios where the profit falls below the (1−α)-quantile of the profit distribution.

subject to:

$$0 \le p_t^{\mathrm{DA}} \le P^{\mathrm{nom}} \quad \forall t = \{1, 2, ..., 24\}$$

$$\Delta_{t\omega} = p_{t\omega}^{\mathrm{real}} - p_t^{\mathrm{DA}} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^{\uparrow} - \Delta_{t\omega}^{\downarrow} \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\uparrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\Delta_{t\omega}^{\downarrow} \ge 0 \quad \forall t = \{1, 2, ..., 24\},\ \forall \omega = \{1, 2, ..., |\omega|\}$$

$$\eta_\omega \ge 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

$$-\sum_{t=1}^{24} \left( \lambda_{t\omega}^{\mathrm{DA}}\ p_t^{\mathrm{DA}} + 0.9\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\uparrow} - 1.2\ \lambda_{t\omega}^{\mathrm{DA}}\ \Delta_{t\omega}^{\downarrow} \right) + \zeta - \eta_\omega \le 0 \quad \forall \omega = \{1, 2, ..., |\omega|\}$$

# Thanks for your attention!

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